Finite-Source Cosmology from Observer Patch Holography

Source, Transfer, and Likelihood Boundaries

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Abstract

A supplied layered event population represents a spatially flat expanding background through its count measure: the scale factor contributes a fourth power per comoving cell and conformal tick. Count ratios recover redshift for congruent constant-profile windows, or in a shrinking-window limit. We connect this exact construction to conditional primordial source models. A specified collar release fixes a screen amplitude; a specified generator density fixes the tilt; physical dilation or complete radial cross-covariances can determine a three-dimensional spectrum. Native load records provide absolute covariance controls but no geometric curvature identification. The source, stress, clock, radial observation and photon-transfer maps must therefore be supplied on the same physical process. A numerical value of the pixel parameter does not provide those maps. Retrospective cosmological comparisons test the declared branches, not a completed source-derived cosmology. Lattice ensembles of the model's link action place the electroweak crossover at 161 GeV, 9×10^{-12} s after the first tick, and show colour deconfinement and the binding of quarks into hadrons.

1 Finite-Source Interface

Let X_r denote a finite observer-screen state at refinement level r. A cosmological source map must emit

$$\mathsf{S}_r(X_r) = (g_{\mu\nu}, T_{\mu\nu}, \mathcal{I}, \mathcal{B})_r, \tag{1}$$

where $T_{\mu\nu}$ is conserved, $\mathcal{I}$ contains gauge-invariant initial data, and $\mathcal{B}$ states the background and boundary conditions. The map must be defined without using the observed sky quantities that its outputs are compared with.

The complete one-shell covariance determines the three-dimensional correlation only for chord lengths $0 \le s \le 2R_\star$. The resulting radial kernel is infinite-dimensional and persists under positivity. Two sufficient routes for a source-derived lift are a scale-natural embedding that transports refinement to physical log-wavenumber dilation and complete radial cross-covariances supporting spherical-Hankel tomography. Each route has its own hypotheses. The physical source map must also specify the radial measure, null-space treatment, clock, normalization, and non-fitting forward residual. No finite source construction satisfying these conditions is supplied here. Its source-dependency graph must be finite and acyclic. This graph is a scientific prerequisite graph among

stress, clock, scale, radial, transfer, and likelihood objects. It is distinct from the semantic read-from order of the consensus layer. If its nodes are instantiated by semantic commits, a declared mapping and complete certified read supports can make the generated informational order verify their execution dependencies. The provenance theorem alone derives neither these scientific nodes nor their prerequisite edges. The rank-three source Gram metric is a finite readout of one carrier's slow response, not an extended space; on the refining source screens the rank-three band leaves bounded-energy port modes at order 2^{-l}, and native dependence crosses at most one seam per ordered repair event, which fixes no laboratory speed or continuum light cone [4, 5, 12]. A cone-selection theorem forces a Lorentzian cone under operational icosahedral and boost covariance of histories, interventions and clock readouts, and the conservative Fibonacci-record family realizes a flat $1+3$ causal-order, count-volume and count-clock limit for a specified population and read law, with ordering fraction of interior timelike intervals tending to $1/10$ [2]. Cosmology requires a common physical realization of those events and clocks with expanding geometry, conserved stress, a primordial source, a radial lift and photon transfer; smooth Einstein promotion requires same-family tensor-curvature convergence or the continuum small-ball/null-balance identification.

The following finite construction realizes a supplied spatially flat expansion profile by weighting the same comoving read order. Its conformal ticks and count ratios are model coordinates; a physical clock and a law selecting the profile are additional inputs.

1.1 Expansion as record density on the source order

The spatially flat Friedmann–Lemaître–Robertson–Walker metric in conformal time η is

$$ds^2 = \sigma(\eta)^2\big(c^2\,d\eta^2 - |dx|^2\big), \tag{2}$$

with scale factor $\sigma(\eta) > 0$; the letter a is the edge radius of the read law. This metric is a positive conformal factor times the flat metric in the coordinates (η, x). A positive conformal factor leaves the causal order of a Lorentzian metric unchanged, and the causal order fixes the metric up to such a factor (Hawking, King and McCarthy 1976; Malament 1977). On the spatially flat family the conformal factor is the entire background, so every profile σ has the causal order of flat space in comoving coordinates. The layered read law of the source-net construction places an event at every (j, s) for a site s in a population S, lets $(j+1, t)$ read (j, s) whenever $\|t - s\| \le a$, and takes the reflexive transitive closure of the reads as the order. Written in comoving coordinates with the tick $\Delta = a/c$ in conformal time, this law generates one finite order for every profile. The profile enters only through the count measure.

Proposition 1.1 (Conformal invariance of the layered order)**.** *Let S be a subset of a real normed space, $a > 0$ an edge radius and $\sigma_j > 0$ the scale at layer j. The physical position of the site s at layer j is $\sigma_j s$ and the physical read radius at that layer is $\sigma_j a$. The physical read from (j, s) to $(j+1, t)$,*

$$\|\sigma_{j+1} t - \sigma_{j+1} s\| \le \sigma_{j+1} a,$$

holds exactly when $\|t - s\| \le a$. The layered orders generated by the physical reads and by the comoving reads coincide, for every profile $(\sigma_j)_{j\ge 0}$.

Proof. For $\sigma > 0$, $\|\sigma t - \sigma s\| = \sigma\|t - s\|$, and multiplication by σ preserves and reflects the comparison with σa. The two read relations are equal, hence so are their reflexive transitive closures. □

Proposition 1.2 (FLRW count measure)**.** *Assign a comoving cell of volume v_s to every site s and the conformal tick Δ to every layer. Under a constant physical event density ρ, the physical tick at layer j is $\sigma_j\Delta$ and the physical cell volume is $\sigma_j^3 v_s$, so a finite set I of events carries the mass*

$$M_\sigma(I) = \rho \sum_{(j,s)\in I} (\sigma_j\Delta)(\sigma_j^3 v_s) = \rho \sum_{(j,s)\in I} \sigma_j^4\, \Delta\, v_s, \tag{3}$$

the discrete form of the four-volume weight $\sigma^4\, d\eta\, d^3x$. Write $M_1(I) = \rho\Delta \sum_{(j,s)\in I} v_s$ for the flat mass. If $0 \le \sigma_{\min} \le \sigma_j \le \sigma_{\max}$ on the layers of I, then

$$\sigma_{\min}^4 M_1(I) \le M_\sigma(I) \le \sigma_{\max}^4 M_1(I). \tag{4}$$

A constant profile $\sigma_j = \sigma$ gives $M_\sigma(I) = \sigma^4 M_1(I)$.

Proof. The identity in (3) is the product of the two scaled factors. Fourth powers are monotone on nonnegative reals and $\Delta v_s \ge 0$, so each term of the sum lies between the same term with σ_j replaced by $\sigma_{\min}$ and by $\sigma_{\max}$. Summing proves (4); the constant case is the common factor σ^4. □

Corollary 1.3 (Count clock in an expanding family)**.** *Let I and J be finite event sets with uniform cells, J nonempty, with $\sigma^I_{\min} \le \sigma_j \le \sigma^I_{\max}$ on the layers of I, $\sigma^J_{\min} \le \sigma_j \le \sigma^J_{\max}$ on the layers of J, $\sigma^I_{\min} \ge 0$ and $\sigma^J_{\min} > 0$. Then*

$$\frac{\sigma^I_{\min}}{\sigma^J_{\max}} \left(\frac{|I|}{|J|}\right)^{1/4} \le \left(\frac{M_\sigma(I)}{M_\sigma(J)}\right)^{1/4} \le \frac{\sigma^I_{\max}}{\sigma^J_{\min}} \left(\frac{|I|}{|J|}\right)^{1/4}. \tag{5}$$

Let I and J approximate comoving diamonds of conformal durations T_I and T_J in the sense of the count-clock theorem for the flat order, so that $(|I|/|J|)^{1/4} \to T_I/T_J$ under refinement. A comoving worldline through a diamond of conformal duration T accrues proper time between $\sigma_{\min}T$ and $\sigma_{\max}T$. In the refinement limit the clock read from the physical masses, $(M_\sigma(I)/M_\sigma(J))^{1/4}$, and the proper-time ratio τ_I/τ_J lie in one interval whose endpoints have the quotient $(\sigma^I_{\max}/\sigma^I_{\min})(\sigma^J_{\max}/\sigma^J_{\min})$; the two readings agree up to this factor in either direction. For a profile continuous in η and diamonds shrinking to points, the factor tends to one and the clock reads the proper-time ratio exactly in the limit.

Proof. Divide the two sandwich inequalities (4) for I and J, with $M_1(I)/M_1(J) = |I|/|J|$ for uniform cells, and take the increasing fourth root; the fourth root of σ^4 is σ. The proper time along the comoving worldline is $\int \sigma\, d\eta$ over the conformal duration, bounded by the extreme scales times T; the ratio of two such enclosures has the stated endpoint quotient. Continuity of σ forces $\sigma_{\max}/\sigma_{\min} \to 1$ on shrinking diamonds. □

Proposition 1.4 (Finite expanding diamonds need not read proper time)**.** *For the supplied radiation profile $\sigma(\eta) = 1 + \eta$ and comoving vertical diamonds starting at $\eta = 0$, take conformal durations 1 and $1/2$. Their continuum mass ratio is*

$$\frac{M_\sigma(1)}{M_\sigma(1/2)} = \frac{1516288}{44451} \ne \left(\frac{12}{5}\right)^4 = \left(\frac{\tau(1)}{\tau(1/2)}\right)^4.$$

Thus their fourth-root volume clock differs from the proper-time ratio even after arbitrarily fine discretization at these fixed durations.

Proof. The common factor $4\pi\rho c^3/3$ cancels from the two integrals

$$M_\sigma(T) \;\propto\; \int_0^{T/2} (1+\eta)^4 \eta^3 \, d\eta + \int_{T/2}^{T} (1+\eta)^4 (T-\eta)^3 \, d\eta.$$

Expanding the polynomials and integrating gives the displayed rational ratio. Along the comoving worldline, $\tau(T) = \int_0^T (1+\eta)\, d\eta = T + T^2/2$, hence $\tau(1)/\tau(1/2) = 12/5$. Subtracting fourth powers gives the strictly positive rational $25944064/27781875$. □

The Lean module `SourceNetClockSeparation` checks the evaluated mass polynomial and rational mismatch; the identification of that polynomial with the geometric integral is the analytical argument above. The resulting relative clock bias is $+0.696366\%$. It is a continuum finite-window effect, not a discretization error. For the supplied profiles $\sigma(\eta) = (1-\eta/2)^{-1}$ and $\sigma(\eta) = [1+(\sqrt{2}-1)\eta]^2$, independent split quadratures give -1.595321% and $+0.265422\%$ at the same durations. The shrinking-diamond hypothesis in Corollary 1.3 is therefore substantive. The exact polynomial control and numerical receipts accompany the observer-dynamics verifier. No expansion profile is selected by this calculation.

Corollary 1.5 (Redshift as a count ratio)**.** *Let two congruent comoving diamonds, with the same comoving cells and the same conformal ticks, lie at the emission epoch, of scale σ_e, and at the reception epoch, of scale σ_0, the profile being constant across each. Their masses are $n_e = \kappa\sigma_e^4$ and $n_0 = \kappa\sigma_0^4$ with the common factor $\kappa = \rho\Delta \sum_{(j,s)\in I} v_s$, and*

$$1 + z = \frac{\sigma_0}{\sigma_e} = \left(\frac{n_0}{n_e}\right)^{1/4}. \tag{6}$$

Proof. The constant case of Proposition 1.2 gives both masses; κ cancels in the ratio, and the positive fourth root of $(\sigma_0/\sigma_e)^4$ is σ_0/σ_e. □

Remark 1.6 (Expansion history as a record-production law). On this family the scale-factor profile is a statement about record production: a comoving cell of volume v produces $\rho\,\sigma(\eta)^4\,\Delta\, v$ records per conformal tick, and a profile $\sigma(\eta)$ is the same datum as a record-production rate per comoving cell. The de Sitter profile $\sigma(\eta) = -1/(H\eta)$ on $\eta < 0$ solves $d\sigma/d\eta = H\sigma^2$, and its record density per comoving cell is $n(\eta) \propto (H\eta)^{-4}$; the fixed-capacity branch of the corpus, $w = -1$, corresponds to this profile. The Lean module `SourceNetConformalRecordDensity` checks the order invariance, the mass identity and sandwich, the clock enclosure (5), the redshift identity and the de Sitter derivative and density identities over the reals.

Three items lie outside this construction. The source law selecting the profile, the record-production rate per comoving cell, is not derived; the profile is a supplied datum. A reweighted flat order determines at most the conformal factor; on the spatially flat family that factor is the entire background, and spatial curvature $k \neq 0$ changes the comoving spatial metric, which this construction does not treat. No physical clock and no population selection by native repair follow from these identities.

1.2 Conditional source spectrum

On a source-stress-complete uniform-density release cut, the screen field is

$$q_r = \Pi_{\geq 2,r} \left[\frac{1}{3} \log \frac{J_{X,r}}{\bar{J}_{X,r}}\right]. \tag{7}$$

The primitive collar ledger selects a relative-MaxEnt release law before the screen energy is evaluated. It excludes spectra, residuals, likelihoods, fitted amplitudes, and measurement-calibrated proxies. The release energy then fixes

$$A_{q,r} = \frac{1}{d_r}\mathbb{E}_{\nu_r^{\mathrm{rel}}}[q_r^{\mathsf{T}} M_r K_{\theta,r} q_r] = \frac{2E_{q,r}^{\mathrm{src}}}{d_r}. \tag{8}$$

The conformal precision and angular spectrum are

$$K_\theta = \frac{\Gamma(\sqrt{-\Delta_{S^2}+1/4}+3/2+\theta/2)}{\Gamma(\sqrt{-\Delta_{S^2}+1/4}-1/2-\theta/2)}, \tag{9}$$

$$C_\ell^q = A_q \frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)}, \qquad \ell \geq 2. \tag{10}$$

The round-sphere Laplacian $-\Delta_{S^2}$ is a premise of this spectrum. The native continuum generator A of the source screen has energy between fixed multiples of the gradient energy of the flat comparison polyhedron; its identification with $-\Delta_{S^2}$ is not established [5]. A required infinitesimal reserve-generator receipt must supply a full-collar density $P_\star/24$. A separately proved weighted orientation-reversal coarea identity then supplies its source-facing half, so

$$\theta = \frac{P_\star}{48}, \qquad n_s = 1 - \frac{P_\star}{48}, \qquad \kappa_{\mathrm{rep}}^{\mathrm{edge}} = \frac{P_\star}{48(P_\star - \varphi)}. \tag{11}$$

A finite one-step survival probability has exponent $-\log u_q(\log b)/\log b$ and does not supply this infinitesimal density.

On the source-dilation branch, the finite source embedding must satisfy

$$D_{s,r}U_r = U_r R_{s,r}, \qquad C_{\zeta,r} = U_r C_{\mathrm{src},r} U_r^*. \tag{12}$$

For a tilted spectrum on all positive wavenumbers, require convergence on the common compact-$\ln k$ core, with the local dilation residual tending to zero, as specified by the source-orbit-transfer theorem of the radial-lift packet in the spacetime paper [2]. A global uniform covariance norm bound would be incompatible with nonzero tilt and unitary dilation. The local receipt instead gives, on that core,

$$D_s^{-1} C_\zeta D_s = e^{-\theta s} C_\zeta. \tag{13}$$

For a nonzero nonnegative radial covariance, this identity forces

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}, \qquad A_\zeta = \frac{A_q}{\pi^{3/2} Z_q^2 (k_\star R_\star)^\theta} \frac{\Gamma(3/2+\theta/2)}{\Gamma(1+\theta/2)}. \tag{14}$$

The tomography branch obtains uniqueness from complete radial cross-covariances. A finite prior selects a conditional radial continuation. Every finite branch publishes its singular spectrum, null basis, finite-window error, positivity check, and forward residual. Temperature and polarization spectra belong to the transfer interface.

1.3 Native records and geometric identification

For fair balanced repair on a connected weighted graph with p ports and r raised loads, stationary centered loads obey

$$\mathrm{Cov}(z(t), z(0)) = \kappa e^{-|t|L/2}\Pi, \quad \kappa = \frac{r(p-r)}{p(p-1)}, \quad \Pi = I - \mathbf{1}\mathbf{1}^{\mathsf{T}}/p.$$

Here t counts unit-rate seam attempts. A normalized eigenmode of the ordinary average $T^{-1}\int_0^T z(t)\,dt$ has variance $2\kappa(u-1+e^{-u})/u^2$, $u=\lambda T/2$; the fluctuation record $T^{-1/2}\int_0^T z(t)\,dt$ instead has limiting covariance $4\kappa L^+$. Incident pre-attempt records, including null attempts, have the same leading term with correction bounded by $3\kappa/d_{\min}$ [3].

On refining $240\cdot 4^l$-port screens at one-third filling, uniform fixed-number preparation and independent fair seam attempts give joint Gaussian snapshot limits at attempt count $\lfloor 1260\cdot 16^l\tau\rfloor$, along one common subsequence:

$$\operatorname{Cov}(z_\tau(f),z_u(g))=\langle f,e^{-|\tau-u|A}g\rangle.$$

The positive strongly local generator obeys $n/49920\le\lambda_n(A)\le 3n/2$ and has spectral dimension two. These finite-dimensional distribution limits assert neither path-space tightness nor a spatial rate. Fixed-area equal port weights and a common temporal filter of bounded L^1 gain W give mean harmonic pseudo-power at most $\pi W^2/(p-1)$, not a bound on each realization. A deterministic temporal filter has spectral slope $d\log V/d\log\lambda\ge -1$. Moreover the eigenvalue growth excludes identification, up to scale and bounded mass shift, with a smooth compact three-dimensional second-order elliptic operator, or bounded energy congruence with its static Dirichlet boundary action. Neither a physical radial coordinate nor a primordial tilt follows [5, 6].

Two exact controls distinguish a geometric source from a load label. If every repair preserves the chart Γ, a geometric volume $V_i(\Gamma)$ relative to its prepared value has log contrast zero at every history. On the isolated twelve-port graph, normalized load and seam drive covariances are respectively $\Pi/11$ and $L^2/360$. Both have diagonal $1/12$, but antipodal covariances are $-1/132$ and zero, and adjacent covariances are $-1/132$ and $-1/45$. Thus equal local variance, locality and icosahedral symmetry do not select spatial power. Since $L_{ii}=5$, adjacent vertices share two neighbors, and $\operatorname{tr}L^2=360$, these entries follow directly by squaring L. Assigning $J_i/\bar{J}_i=e^{3u_i}$ to a load observable u_i supplies a constitutive rule, not a derived geometric volume. The finite-history proof and exact enumeration are retained in `Geometry.NativeGeometricSourceIdentification` and `code/native_geometric_source` [15]. A cosmological source must supply this metric/record interface as well as its spatial and temporal calibration.

1.4 Exact Fourier control for the golden source population

The deterministic Cartesian population $x_b=\mathcal{L}(\{b_1\varphi\},\{b_2\varphi\},\{b_3\varphi\})$, $0\le b_i<q$, has an exact fixed-cube Fourier response. Here $\varphi=(1+\sqrt{5})/2$, $\mathcal{L}$ is the declared cube size, and the integer mode n has wave vector $2\pi n/\mathcal{L}$. Its normalized amplitude factors as

$$F_q(n)=q^{-3}\sum_b e^{2\pi i n\cdot x_b/\mathcal{L}}=\prod_{j=1}^{3}a_q(n_j),\qquad a_q(0)=1,$$

$$a_q(m)=e^{\pi i m\varphi(q-1)}\frac{\sin(\pi mq\varphi)}{q\sin(\pi m\varphi)}\quad(m\neq 0).$$

This follows by separating the three finite sums, using integer-mode periodicity to remove fractional parts, and evaluating each geometric series. For $q=F_j$, the Fibonacci identity $\varphi F_j-F_{j+1}=(-1)^{j+1}\varphi^{-j}$, with $F_j\varphi^{-j}\to 1/\sqrt{5}$, gives for fixed n with h nonzero coordinates

$$q^{4h}|F_q(n)|^2\longrightarrow\prod_{n_a\neq 0}\frac{(\pi n_a)^2}{5\sin^2(\pi n_a\varphi)}.$$

The denominator is nonzero by irrationality. The small-angle limit of each numerator proves the rate. It concerns fixed integer modes in a fixed cube. It is not an infinite-volume hyperuniformity result or a primordial tilt. Finite directional power ratios can be large while both powers vanish; some finite modes remain above the independent-uniform structure-factor reference. The exact formula, independent small-cloud checks and the retained numerical receipt are archived [16]. No physical length or curvature amplitude is selected by this supplied coordinate population. The Lean module `Geometry.GoldenSourceFourier` [15] checks tensor factorization, fractional-coordinate phase equality and the finite geometric-series identity. The fixed-mode asymptotic argument above is analytical, not a claimed formal limit theorem.

1.5 Common-bulk consistency beyond a single screen

A further sufficient uniqueness route uses an unbounded sequence of round radii, on one common Euclidean time slice and in one fixed normalization. If one homogeneous isotropic generalized bulk covariance has exact screen spectra $C_\ell(R_j) = 2\pi A_j/[\ell(\ell+1)]$ for all $\ell \geq 1$, then overlapping chord intervals force all $A_j = A$, $\xi(s) = -A \log s + c$ almost everywhere, and $P_3(k) = 2\pi^2 A/k^3$ for $k > 0$. The radial theorem assumes well-defined restrictions, a locally integrable off-diagonal covariance and no contact terms. Removing the dipole retains $c + ds^2$ freedom. A finite family of separate screen marginals does not supply the common bulk or unbounded coverage: positive perturbations supported beyond the largest measured chord preserve all those marginals. The detailed radial theorem and its explicit positive-baseline control therefore strengthen the conditional inverse statement without identifying the native load record as curvature. The native screen energy supplies no such common bulk: by the boundary-congruence exclusion of Section 1.3, a bulk reading of the screen needs a radial observation algebra that the native energy does not supply [5].

2 Dark-Sector Interface

The dark-matter paper proposes anomalous modular charge on overlap collars. For a conserved total charge Q_A in a comoving region of physical volume $V_0 a^3$, its homogeneous density satisfies

$$\rho_A = \frac{Q_A}{V_0 a^3} \propto a^{-3}. \tag{15}$$

Under the separate homogeneous continuity equation with no exchange, this dilution is the pressureless background law. Deep-regime scale covariance and quadrature composition characterize the static spherical acceleration law and baryonic Tully–Fisher scaling.

Equation (15) does not determine the abundance, initial perturbations, relativistic stress, gravitational slip, sound speed, nonlinear evolution, or coupling to the visible sector. A cosmological use requires a covariant completion that supplies these quantities from one common source and obeys the full energy-momentum ledger. That completion is not supplied here.

3 Primordial source and conditional comparisons

The source must provide conserved stress, a common clock and release surface, scalar/vector/tensor initial data, radial observation and physical scale, and controlled refinement. Neither the value

of P nor a finite gauge-current algebra determines these objects. In particular, a conditional particle/memory fixed point does not derive the separate cosmological generator density $P/24$.

On the declared density branch, $n_s = 1 - P_\star/48$ gives 0.96602 at the comparison pixel $P_\star = 1.6309682094$, which is located from the measured fine-structure constant. The continuous survival cocycle alone admits every exponent, so this numerical density is a substantive premise. At that value the retrospective position in the ACT DR6, Planck, lensing and DESI 2024 BAO joint posterior is the lower 0.612 percent tail; the same value is within one standard deviation of Planck 2018 and the two cited SPT combinations. These are seen-data comparisons under baseline ΛCDM, not likelihood ratios or a selection of the physical source [14, 13]. The complete comparisons, including background tensions, are given in the likelihood companion [1].

A single-clock Gaussian source has no independent source isocurvature or connected higher Gaussian cumulants, and a fixed dilation exponent has zero running. Tensor zero needs an extra condition: no tensor forcing alone permits homogeneous solutions. For example, $h_k'' + 2h_k'/\tau + k^2 h_k = 0$ admits $h_k = \sin(k\tau)/(k\tau)$. Vanishing initial amplitude and velocity, together with zero forcing, select the zero linear solution. A finite observational upper bound excludes only alternatives above that bound; single-field slow roll has no universal positive tensor floor.

4 Transfer Interface

Given a completed source map, the transfer calculation must declare the background species, equations of state, collision terms, recombination model, initial modes, gauge convention, numerical tolerances, and observable projection. The finite particle comparison computes no particle abundance or early-Universe population distribution, so those background species densities are imported [8]. An equation of state needs its own strain response: identical fixed-volume records admit $H_\alpha(V) = Q(V/V_0)^{-\alpha}$ with $p/\rho = \alpha$, and an additive energy offset changes p/ρ while leaving pressure unchanged [7]. Standard Einstein–Boltzmann software can test this interface with imported inputs. Such a run tests the solver path rather than an OPH source theorem.

The minimum transfer outputs are

$$(C_\ell^{TT}, C_\ell^{TE}, C_\ell^{EE}, C_L^{\phi\phi}, P_m(k,z), H(z), D_A(z), f\sigma_8(z)). \tag{16}$$

All outputs must arise from one compatible background and perturbation system.

5 Likelihood Interface

A data comparison must state the datasets, masks, covariances, nuisance parameters, priors, and combination rule. Overlapping samples and shared calibrations must be represented in the covariance model. Visual overlays, diagonal residual sums, and comparisons made after selecting a formula are diagnostics.

The cosmic microwave background (CMB), galaxy, and background comparisons are diagnostic. These computations supply neither a prospectively fixed cosmological comparison nor a falsification verdict.

6 Thermal history on the model's link action

The executable model has its own unit of time. One tick is one radian of phase at the cell energy, $\hbar/E_{\mathrm{cell}} = 6.885 \times 10^{-44}$ s with $E_{\mathrm{cell}} = E_* P^{-1/2} = 9.56 \times 10^{18}$ GeV. The stored runs cover the first few hundred ticks. In them every defect class appears in pairs on the first consistent slice, class and conjugate stay balanced, and the classes of left-handed quarks, right-handed quarks and leptons take 24.1, 19.6 and 12.7 percent of all defect identities per class, against 21.4, 21.4 and 14.3 percent for equal occupation of the Standard Model states of each class [8].

Later epochs are reached by lattice ensembles whose extent of N_t ticks in the time direction is read as the temperature $T = E_{\mathrm{cell}}/N_t$. In the radiation era with the model's own content, twelve gauge bosons, one Higgs doublet and fifteen gauge-charged fermion states per copy in three copies, the count of relativistic states is $g_* = 106.75$ and the standard relation $t = 2.42\,\mathrm{s}\, g_*^{-1/2} (\mathrm{MeV}/T)^2$ converts a temperature into a time. Three measurements use the model's link action.

- With the couplings of the declared coupling chain and the published reduction to the three-dimensional SU(2) and Higgs theory [9], the electroweak transition is a crossover at $T_c = 161.13 \pm 0.13 \pm 1.6$ GeV, which is 9.0×10^{-12} s or 1.3×10^{32} ticks after the first. The Standard Model lattice value is 159.5 ± 1.5 GeV [10].
- Restricted to colour, the seam cost of the links is the Wilson plaquette action. At $N_t = 4$ the links deconfine in a first-order transition with $T_c/\sqrt{\sigma} = 0.614(18)$; the published value for the same theory at this lattice spacing is 0.598(3) [11].
- Quenched Wilson quarks on the same links bind, with a linearly rising static potential and $m_N/m_\rho = 1.45 \pm 0.13$ against the measured 1.21, and the binding melts above the transition.

The crossover takes the chain's declared couplings as inputs, and the colour and hadron numbers are those of SU(3) gauge theory with the Wilson action, which the model's colour links are once continuous links are declared. A temperature of the colour transition in MeV, the baryon excess and the light-element abundances are not computed.

7 Scope

The count identities and finite Fourier response are established for the supplied population. Screen covariance, amplitude and tilt depend on the specified precision and source law. A physical cosmology additionally needs the common state, metric, stress, radial and transfer identification. The passive native screen does not supply that identification, and a solver using supplied primordial inputs is a conditional forward model.

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